



8086 Interrupts

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✓ **Syllabus Topic : 8086 Interrupts, Types of Interrupts Interrupt Service Routine, Servicing of Interrupts by 8086 Microprocessor**

➔ 3.1 8086 | Interrupts

- An interrupt is a special condition that arises during the working of a μP .
- The μP services it by executing a subroutine called **Interrupt Service Routine (ISR)**.
- There are 3 sources of interrupts for 8086:

🔧 External Signal (Hardware Interrupts)

- These interrupts occur as signals on the external pins of the μP .
- 8086 has two pins to accept hardware interrupts, NMI and INTR.

🔧 Special instructions (Software Interrupts)

- These interrupts are caused by writing the software interrupt instruction **INT n** where "n" can be any value from 0 to 255 (00H to FFH).
- Hence all 256 interrupts can be invoked by software.

🔧 Condition Produced by the Program (Internally Generated Interrupts)

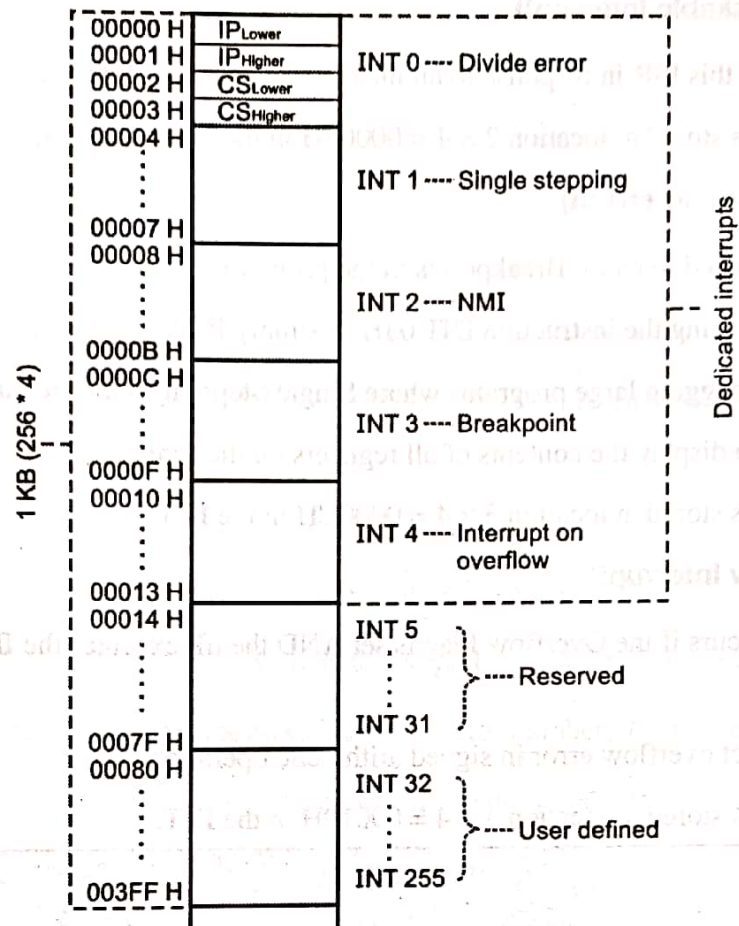
- 8086 is interrupted when some special conditions occur while executing certain instructions in the program.
- Eg: An error in division automatically causes the INT 0 interrupt.

✓ **Syllabus Topic : Interrupt Vector Table (IVT)**

➔ 3.2 Interrupt Vector Table (IVT)

Q. Explain : Types of interrupts (May 17, May 18, Dec. 18, 10 Marks, May 19, 5 Marks)

- The IVT contains ISR address for the 256 interrupts.
- Each ISR address is stored as CS and IP.
- As each ISR address is of 4 bytes (2-CS and 2-IP), each ISR address requires 4 locations to be stored. There are 256 interrupts: INT 0 ... INT 255 $\therefore$ the total size of the IVT is $256 \times 4 = 1\text{KB}$.
- The first 1KB of memory, address 00000 H ... 003FF H, are reserved for the IVT.
- Whenever an interrupt INT N occurs, μP does $N \times 4$ to get values of IP and CS from the IVT and hence perform the ISR.





3. INT 2 (Non Maskable Interrupt)

- The μP executes this ISR in response to an interrupt on the NMI line.
- Its ISR address is stored at location $2 \times 4 = 00008H$ in the IVT.

4. INT 3 (Breakpoint Interrupt)

- This interrupt is used to cause Breakpoints in the program.
- It is caused by writing the instruction INT 03H or simply INT.
- It is useful in debugging large programs where Single Stepping is inefficient.
- Its ISR is used to display the contents of all registers on the screen.
- Its ISR address is stored at location $3 \times 4 = 0000CH$ in the IVT.

5. INT 4 (Overflow Interrupt)

- This interrupt occurs if the Overflow Flag is set AND the μP executes the INTO instruction (Interrupt on overflow).
- It is used to detect overflow error in signed arithmetic operations.
- Its ISR address is stored at location $4 \times 4 = 00010H$ in the IVT.

Please Note :

INT 0 ... INT 4 are called dedicated interrupts as these interrupts are dedicated for the above-mentioned special conditions.

3.2.2 Reserved Interrupts

INT 5 ... INT 31

- These levels are reserved by INTEL to be used in higher processors like 80386, Pentium etc.
- They are not available to the user.

3.2.3 User defined Interrupts

INT 32 ... INT 255

- These are user defined, software interrupts.
- ISRs for these interrupts are written by the users to service various user defined conditions.
- These interrupts are invoked by writing the instruction INT n.
- Its ISR address is obtained by the μP from location $n \times 4$ in the IVT.

3.2.4 Hardware Interrupts

1. NMI (Non Maskable Interrupt)

- This is a non-maskable, edge triggered, high priority interrupt.
- On receiving an interrupt on NMI line, the μP executes INT 2.
- μP obtains the ISR address from location $2 \times 4 = 00008H$ from the IVT.
- It reads 4 locations starting from this address to get the values for IP and CS, to execute the ISR.

2. INTR

- This is a maskable, level triggered, low priority interrupt.
- On receiving an interrupt on INTR line, the μP executes 2 $\overline{INTA}$ pulses.

1st $\overline{INTA}$ pulse --- the interrupting device calculates (prepares to send) the vector number.

2nd $\overline{INTA}$ pulse --- the interrupting device sends the vector number "N" to the μP .

Now μP multiplies $N \times 4$ and goes to the corresponding location in the IVT to obtain the ISR address.

- INTR is a maskable interrupt.
- It is masked by making $IF = 0$ by software through CLI instruction.
- It is unmasked by making $IF = 1$ by software through STI instruction.

3.3 How 8086 Responds to an Interrupt

Steps Performed when an Interrupt "INT N" Occurs

1. The μP will PUSH Flag register into the Stack.

$SS:[SP-1], SS:[SP-2] \leftarrow \text{Flag}$

$SP \leftarrow SP - 2$

2. Clear IF and TF in the Flag register and thus disables INTR interrupt.

$IF \leftarrow 0, TF \leftarrow 0$

3. PUSH CS into the Stack

$SS : [SP-1], SS:[SP-2] \leftarrow CS$

$SP \leftarrow SP - 2$

**4. PUSH IP into the Stack**

$$SS:[SP-1], SS:[SP-2] \leftarrow IP$$

$$SP \leftarrow SP - 2$$
5. Load new IP from the IVT

$$IP \leftarrow [N \times 4], [N \times 4 + 1]$$
6. Load new CS from the IVT

$$IP \leftarrow [N \times 4 + 2], [N \times 4 + 3]$$

- Since CS and IP get new values, control shifts to the address of the ISR and the ISR thus begins.
- At the end of the ISR the μP encounters the IRET instruction and returns to the main program in the following steps.

Steps performed to return to main program using IRET instruction

1. The μP will restore IP from the stack

$$IP \leftarrow SS:[SP], SS:[SP+1]$$

$$SP \leftarrow SP + 2$$

2. The μP will restore CS from the stack

$$CS \leftarrow SS:[SP], SS:[SP+1]$$

$$SP \leftarrow SP + 2$$

3. The μP will restore FLAG register from the stack

$$Flag \leftarrow SS:[SP], SS:[SP+1]$$

$$SP \leftarrow SP + 2$$
3.4 Interrupt Priorities

Divide Error, INT n, INTO	(Simultaneous occurrence)	(To interrupt another ISR)
Divide Error, INT n, INTO	1 (Highest)	Can interrupt any ISR
NMI	2	
INTR	3	Cannot interrupt an ISR (IF, TF $\leftarrow$ 0)
Single Stepping	4 (Lowest)	



Priority in 8086 interrupts is of two types :

1. Simultaneous Occurrence
2. Ability to interrupt another ISR

1. Simultaneous Occurrence

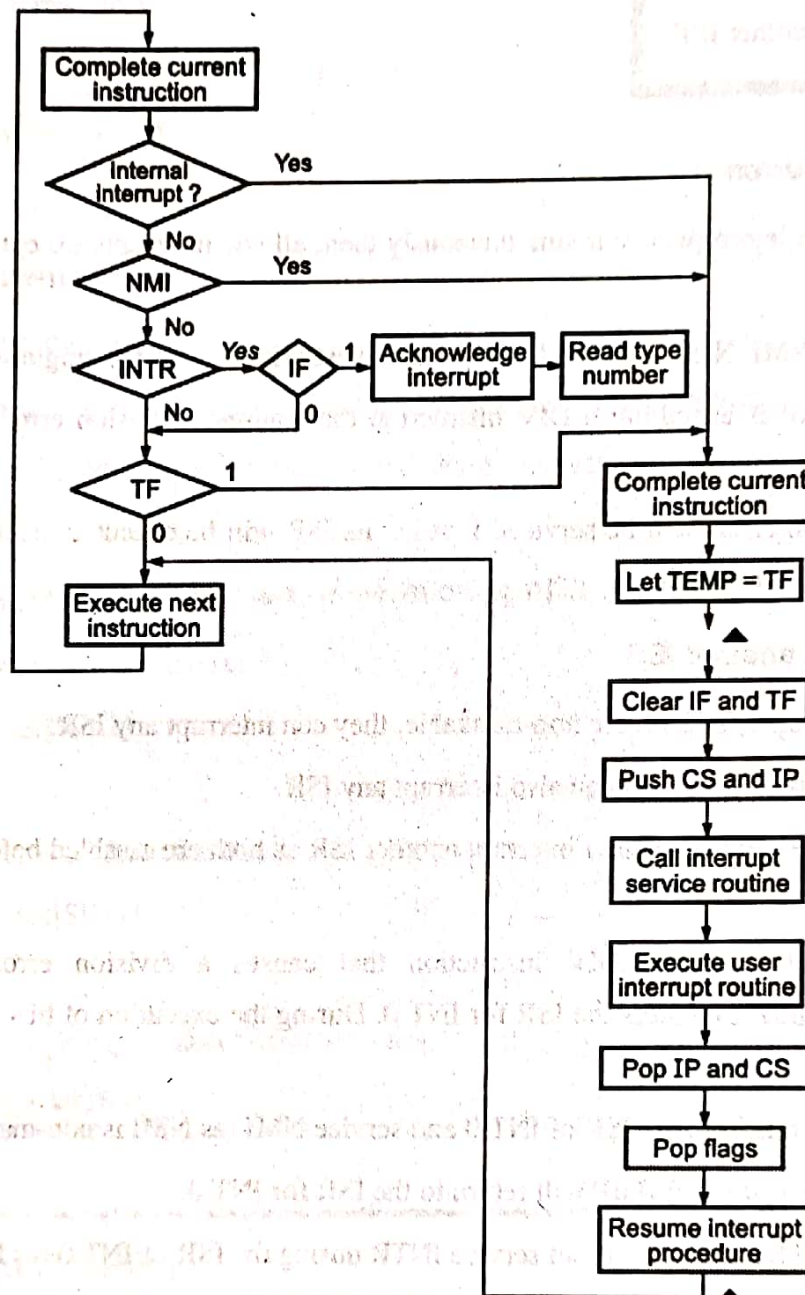
- When more than one interrupts occur simultaneously then, all s/w interrupts except single stepping, get the highest priority.
- This is followed by NMI. Next is INTR. Finally, the lowest priority is of the single stepping interrupt.
- E.g. : Assume the μP is executing a DIV instruction that causes a division error and simultaneously INTR occurs.
- Here INT 0 (Division error) will be serviced first i.e. its ISR will be executed, as it has higher priority, and then INTR will be serviced.

2. Ability to interrupt another ISR

- Since software interrupts (INT N) are non-maskable, they can interrupt any ISR.
- NMI is also non-maskable hence it can also interrupt any ISR.
- But INTR and Single stepping cannot interrupt another ISR as both are disabled before μP enters an ISR by $IF \leftarrow 0$ and $TF \leftarrow 0$.
- Eg: Assume the μP executes DIV instruction that causes a division error. So μP gets the INT 0 interrupt and now μP enters the ISR for INT 0. During the execution of this ISR, NMI and INTR occur.
- Here μP will branch out from the ISR of INT 0 and service NMI (as NMI is non-maskable).
- After completing the ISR of NMI μP will return to the ISR for INT 0.
- INTR is still pending but the μP will not service INTR during the ISR of INT 0 (as $IF \leftarrow 0$).
- μP will first finish the INT 0 ISR and only then service INTR.
- Thus INTR and Single stepping cannot interrupt an existing ISR.



3.5 Interrupt Priority Flowchart



(1A25) Fig. 3.5.1 : Flow chart for interrupt processing



3.6 University Questions and Answers

⇒ May 17

Q. 5(b) Explain interrupt structure of 8086. (Refer section 3.2)

(10 Marks)

⇒ May 18

Q. 2(a) Explain interrupt structure of 8086. (Refer section 3.2)

(10 Marks)

⇒ Dec. 18

Q. 5(b) Explain interrupt structure of 8086. (Refer section 3.2)

(10 Marks)

⇒ May 19

Q. 6(b)(i) Explain : Types of interrupts (Refer section 3.2)

(5 Marks)

Chapter ends...

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